Course code	Course title	L	T	P	J	C
CSE1001	Problem Solving and Programming	3	0	2	0	4

Pre-requisite NIL Syllabus version 1.0

Course Objectives:

1. To understand the basic concepts of problem solving using algorithms and flowcharts.
2. To develop programs using control structures and functions in a high level language.
3. To apply arrays, strings and pointers to solve engineering problems.

Course Outcomes:

On completion of this course, the student will be able to:
1. Explain the steps in problem solving and represent solutions using algorithms and flowcharts.
2. Write programs using decision making statements, loops and functions.
3. Develop programs that process arrays and strings.
4. Use pointers and structures to build modular programs.

Module:1 **Introduction to Problem Solving** 6 hours

Problem solving strategies, algorithms, flowcharts, pseudocode, structure of a program, variables, data types, operators and expressions.

Module:2 **Control Structures and Functions** 9 hours

Decision making statements, loops, nested loops, break and continue, functions, parameter passing, recursion.

Module:3 **Arrays and Strings** 8 hours

One and two dimensional arrays, searching and sorting, string handling functions, array of strings.

Module:4 **Pointers and Structures** 8 hours

Pointers, pointer arithmetic, dynamic memory allocation, structures, unions, enumerations.

Module:5 **Files** 8 hours

File operations, sequential and random access files, command line arguments.

Module:6 **Contemporary Issues** 2 hours

Guest lecture by industry experts.

Total Lecture hours: 45 hours

Text Book(s)

1. Yashavant Kanetkar, Let Us C, 16th Edition, BPB Publications, 2019.
2. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, 2nd Edition, Pearson, 2015.

Reference Books

1. Herbert Schildt, C: The Complete Reference, 4th Edition, McGraw Hill, 2017.
2. E. Balagurusamy, Programming in ANSI C, 8th Edition, McGraw Hill, 2019.

Mode of Evaluation: CAT, Assignment, Quiz, FAT

List of Challenging Experiments (Indicative)

1. Programs using input and output statements and arithmetic operators.
2. Programs on decision making statements.
3. Programs using loops: pattern printing, series generation.
4. Programs on one dimensional and two dimensional arrays.
5. String manipulation programs without using library functions.
6. Programs using pointers and dynamic memory allocation.

Total Laboratory Hours 30 hours

Recommended by Board of Studies 12-08-2022
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